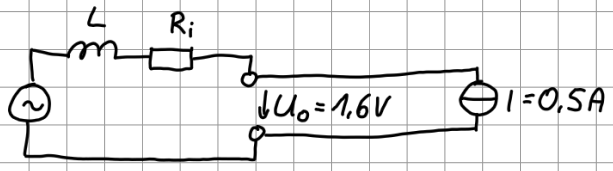


Shimano DH-T4050-1D

4-Leiter-Messung:

Im Stillstand 500mA einspeisen:



$Z_{ges} = \sqrt{X_L^2 + R^2}$ mit $X_L = \omega \cdot L$ und $\omega = 0 \rightarrow Z_{ges} = R$

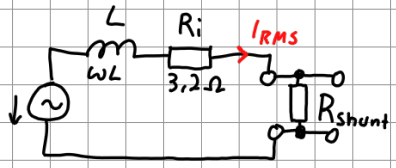
$R = \frac{U_0}{I} = \frac{1,6V}{0,5A} = 3,2\,\Omega$

Leerlaufspannung:

bei f = 26Hz: $V_{RMS} = 8,78V = V_0$

Shunt Messung:

bei f = 26 Hz: $V_{RMS} = 136mV$ $I_{RMS} = \frac{V_{RMS}}{R_{shunt}} = \frac{0,136V}{0,3\Omega} = 453mA$



$Z_{ges} = \frac{V_0}{I_{RMS}} = \frac{8,78V}{0,453A} = 19,38\,\Omega$

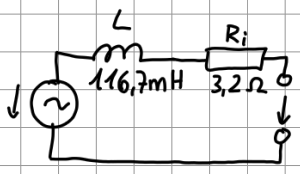
$Z_{ges}^2 = X_L^2 + (R_i + R_s)^2$

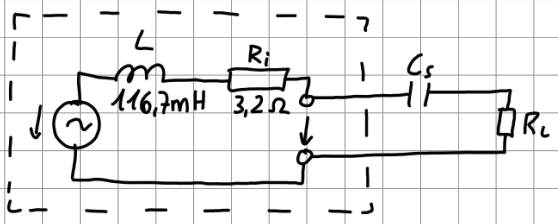
$X_L = \sqrt{Z_{ges}^2 - (R_i + R_s)^2}$

$X_L = 19,06\,\Omega$

$L = \frac{X_L}{\omega} = \frac{19,06\,\Omega}{2 \cdot \pi \cdot 26Hz} = 116,7mH$

ESB mit Werten





$$C = \frac{1}{\omega^2 \cdot L} = \frac{1}{(2\pi \cdot f)^2 \cdot 116{,}7\text{mH}} = 297\,\mu\text{F}$$

$$\frac{15}{27} = \frac{5}{9}\ \frac{\text{kmh}}{\text{Hz}}$$